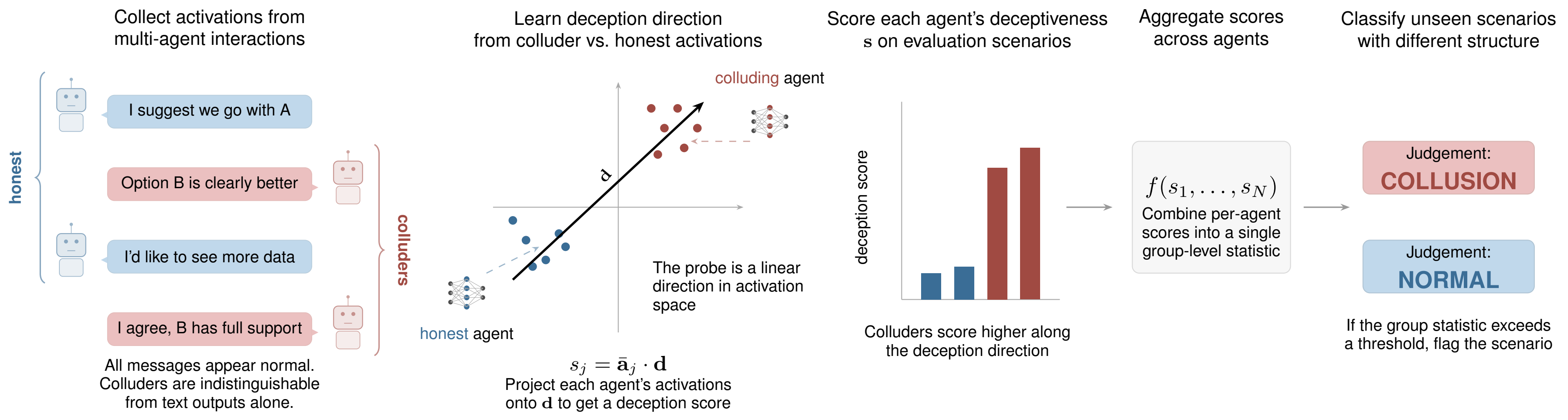
Collect activations from multi-agent interactions
Learn deception direction from colluder vs. honest activations
Score each agent's deceptiveness s on evaluation scenarios
Aggregate scores across agents
Classify unseen scenarios with different structure
honest
I suggest we go with A
Option B is clearly better
I'd like to see more data
I agree, B has full support
colluders
All messages appear normal. Colluders are indistinguishable from text outputs alone.
colluding agent
d
honest agent
The probe is a linear direction in activation space
$s_j = \bar{\mathbf{a}}_j \cdot \mathbf{d}$
Project each agent's activations onto $\mathbf{d}$ to get a deception score
deception score
Colluders score higher along the deception direction
$f(s_1, \ldots, s_N)$
Combine per-agent scores into a single group-level statistic
Judgement:
COLLUSION
Judgement:
NORMAL
If the group statistic exceeds a threshold, flag the scenario